

```
In [1]: class ClassroomAgent:

    def __init__(self):
        self.previous_mode = None

    def act(self, temp, occupied):

        if not occupied:
            target_mode = "ECO"
        elif temp > 26:
            target_mode = "COOL"
        elif temp < 20:
            target_mode = "WARM"
        else:
            target_mode = "IDLE"

        if target_mode == self.previous_mode:
            command = False
        else:
            command = True

        old_mode = self.previous_mode
        self.previous_mode = target_mode

        return old_mode, target_mode, command

agent = ClassroomAgent()

percepts = [
    (29, True),
    (29, True),
    (20, True),
    (26, True),
    (19, True),
    (29, False),
    (29, False)
]

print("Previous Mode | Percept | Target | Command")

for percept in percepts:
    previous, target, command = agent.act(*percept)

    if command:
        result = "SEND COMMAND"
    else:
        result = "NO COMMAND"

    print(previous, "|", percept, "|", target, "|", result)
```

Previous Mode	Percept	Target	Command
None	(29, True)	COOL	SEND COMMAND
COOL	(29, True)	COOL	NO COMMAND
COOL	(20, True)	IDLE	SEND COMMAND
IDLE	(26, True)	IDLE	NO COMMAND
IDLE	(19, True)	WARM	SEND COMMAND
WARM	(29, False)	ECO	SEND COMMAND
ECO	(29, False)	ECO	NO COMMAND

In []: